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LAB 3 — BUILD CIRCUITS WITH CIRCUIT BUILDER

For

Technical Drawing with AutoCAD Electrical

TGS Ref No: TGS-2023037466

Conducted by

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Version 6.0

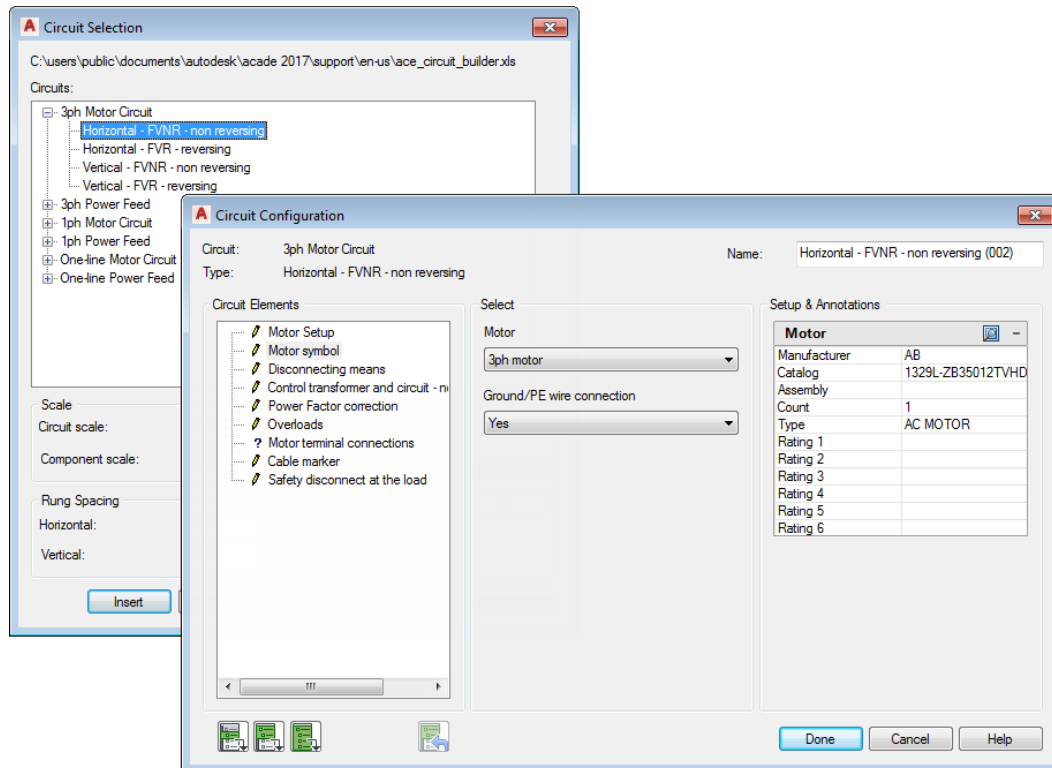
Lab 3 — Build Circuits with Circuit Builder

Topic 02: Analysing Technical Drawings

Maps to: A4 — analyse drawings to determine construction needs; assemble standard motor-control circuits

Goal

Use Circuit Builder to insert a three-phase motor-control circuit built dynamically from your selections, then save a frequently used circuit and re-insert it from the icon menu.



What you'll build

A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block. (Tools: Circuit Builder · Icon Menu · Schematic components · Saved circuits.)

Starter files (in this folder)

- 2-0_Blank.dwg — source: github.com/PacktPublishing/AutoCAD-2025-Best-Practices-Tips-and-Techniques

Step-by-step

1. Open the supplied blank sheet 2-0_Blank.dwg inside your TRAINING project and insert a ladder to carry the circuit.

AE LADDER

2. Launch Circuit Builder from the Schematic tab and pick a 3-phase motor circuit from the tree.

AECIRCBUILDER

3. Step through the Circuit Configuration dialog — select the disconnect, protection and control options; the circuit is built dynamically from your selections.

Circuit Builder → Configure → Insert

4. Insert a push button and a selector switch from the Icon Menu to complete the control rung.

AECOMPONENT (Icon Menu)

5. Select the finished circuit and save it with Save Circuit to Icon Menu for reuse.

Schematic → Circuit Clipboard → Save Circuit

6. Insert the saved circuit onto a clear area of the sheet and watch tags update to stay unique.

Icon Menu → Saved Circuits

Test it

The motor circuit sits on the ladder with unique component tags (M1, PB1 ...), and your saved circuit re-inserts with automatically retagged components — construction needs are met without redrawing.